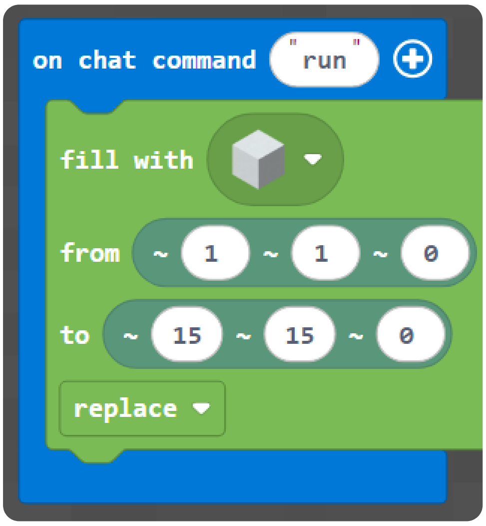
🧊 M003 Week 3 — English Worksheet (Intermediate)

Topic: Pixel Art on the Wall (x, y) · **Course:** 3D Coordinates · **Level:** Intermediate · **Time:** about 40 minutes

This week you **copy a picture made of blocks** onto your wall. Every block sits at a spot named by **two numbers — (x, y)**. x is how far **across**, y is how far **up**. You read the picture, find each block's (x, y), and place it.

First, build your canvas. Run this to make a blank **15 × 15** wall, then put a **red block at your feet** as your **home spot**:

🧩 Blocks



🐍 Python

```
def on_run():
    blocks.fill(WHITE_CONCRETE,
        pos(1, 1, 0),
        pos(15, 15, 0),
        FillOperation.REPLACE)
player.on_chat("run", on_run)
```

> 🔴 **Big idea:** (x, y) is counted from your **home spot** — not from the corner of the world. Stand on the red block *every* time you run, or the whole picture shifts. Move your feet, move your picture.

1 · Predict 🔮

Read each set of steps. Write what you think you will see — name the colour, where it lands, and the shape it makes.

```
place red block at (5, 1)
place red block at (6, 1)
place red block at (7, 1)
```

What do these three blocks make? Across or up? Which number stays the same?

```
place green block at (8, 13)
place green block at (8, 14)
place green block at (8, 15)
```

These three share x = 8. Near the bottom of the wall or near the top? Across or up?

```
place yellow block at (6, 1)
place yellow block at (7, 2)
place yellow block at (8, 3)
```

Both numbers change each time. Does this make a straight row, a straight column, or a stair going up?

2 · Spot the Bug 🐛

Each block of code was meant to do something, but it is broken. Read what it is **supposed** to do, then rewrite it so it works. After that, explain why the original was wrong and why your fix works.

Bug A — These should make the **bottom row of the pineapple**: five yellow blocks from (6, 1) to (10, 1). Right now they all land in the **same spot**.

```
place yellow block at (6, 1)
place yellow block at (6, 1)
place yellow block at (6, 1)
place yellow block at (6, 1)
place yellow block at (6, 1)
```

Hint: to move across, the **x** has to change each time.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — The apple stem should go **up** at x = 8, at (8, 13), (8, 14), (8, 15). Right now it grows **across** instead.

```
place brown block at (13, 8)
place brown block at (14, 8)
place brown block at (15, 8)
```

Hint: check which slot the changing number is in — the x slot (across) or the y slot (up). The two numbers got swapped.

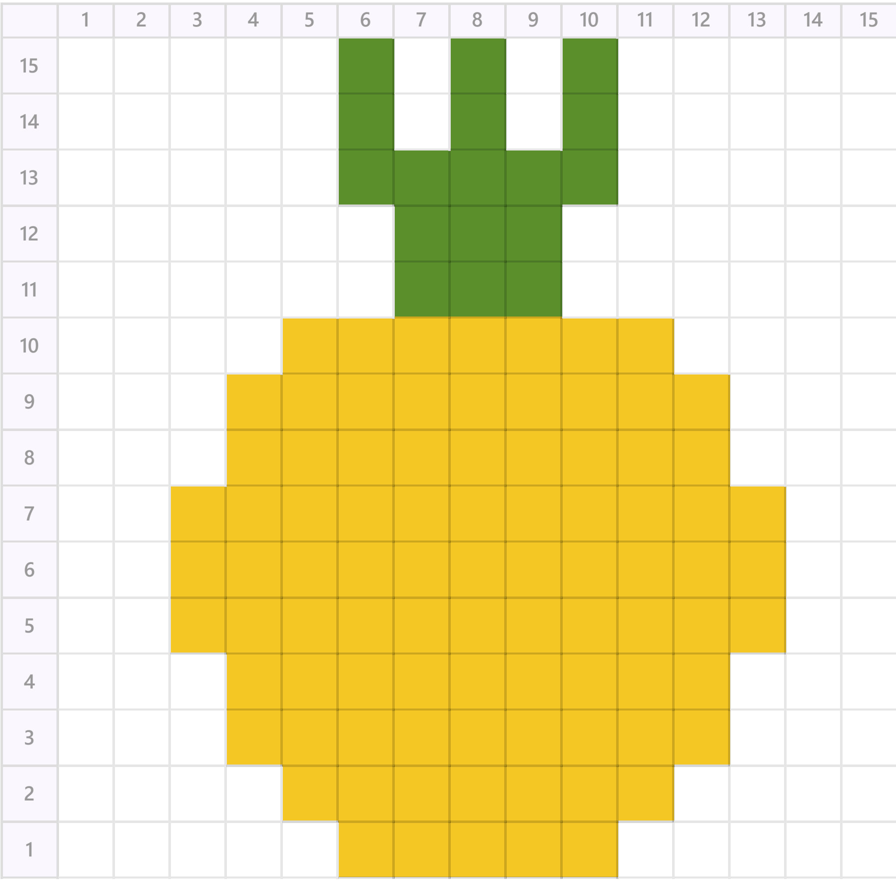
Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Tell Me What You Built 📸

Now switch to your homework world and stand on your **home spot** (the red block). Copy the **pineapple** below onto your wall, block by block, by reading the (x, y) of every colored square. The numbers across the **top** are **x**, the numbers down the **side** are **y**.

🍍 Pineapple



When your picture is finished, send a photo of it, then explain what you did. Use these sentence starters — write 4 to 6 sentences total.

The lowest block in my pineapple is at (x = …, y = …).

To build a row going across, I changed the … number each time.

To build the leaves going up, I changed the … number.

The trickiest part to read off the grid was …

One time my picture landed in the wrong place because …

If I had more time, I would …

4 · Record Your Walkthrough 🎥

Now take a video on your phone (or a parent's phone) while you show your pixel art on the wall. Talk through it like you are teaching someone who has never seen it. Try to use these words: **x**, **y**, **coordinate**, **across**, **up**, **home spot**.

1. Show your home spot (the red block) and say why you stand on it every time.
2. Show your finished pineapple from the front.
3. Point at one block and read its coordinate out loud: "this one is at x ..., y ...".
4. Show one block that you placed in the wrong spot at first, and how you fixed it.

Write what you will say in your video. Use the space below to plan it before you record — you can read from it while filming.

Submit ✅

Send this worksheet + your walkthrough video to teacher on KakaoTalk.